

Speech Emotion Recognition (SER) – Project Specification

Course: Machine Learning

Difficulty: Intermediate

Type: Individual Project

Due: [Set your date]

Objective

Build a Speech Emotion Recognition (SER) system that classifies human emotions from audio recordings using Librosa for feature extraction and MLPClassifier from Scikit-learn.

Students will learn:

- Audio preprocessing
- Feature engineering
- Classical ML modeling
- Evaluation & reporting

Project Requirements

1. Dataset

Use any of the following public speech emotion datasets (choose at least one):

- RAVDESS
- TESS
- CREMA-D
- EMO-DB

You must:

- Download and organize the dataset
- Have at least 5 different emotion classes

2. Audio Preprocessing & Feature Extraction (Mandatory)

Using Librosa, extract the following features from each audio file:

- MFCCs (20 coefficients minimum)
- Chroma features
- Mel Spectrogram
- Zero-Crossing Rate
- Spectral Centroid

Combine all extracted features into one feature vector per audio file.

3. Model Training (MLPClassifier)

You must:

- Split dataset into train/test

- Scale features with StandardScaler
- Train an MLPClassifier
- Tune at least 3 hyperparameters (hidden_layer_sizes, learning_rate, alpha, etc.)

4. Model Evaluation

Your report must include:

- Accuracy score
- Classification report
- Confusion matrix
- Discussion of results & errors

5. Deliverables

You must submit:

A. Source Code

- Python script or Notebook with feature extraction, training, and evaluation

B. Saved Model

- Export trained MLP model as .pkl or .joblib

C. Short Report (2–4 pages)

- Dataset description
- Features used
- Model architecture
- Results & graphs
- What could be improved

D. (Optional for Bonus)

- Streamlit web app for uploading audio and predicting emotion